INDIAN INSTITUTE OF TECHNOLOGY MADRAS
CHENNAI 600 036

IDDD Curriculum

INDIAN INSTITUTE OF TECHNOLOGY MADRAS

Curriculum for IDDD Programme

Inter Disciplinary Dual Degree Programme
Category and Branch-wise credit requirements

Applicable for the following ID-DD programmes

- **Advanced Materials and Nano Technology-157 cr**
- **Biomedical Engineering-157**
- **Computational Engineering-157**
- **Data Science-157**
- **Energy Systems-157**
- **Cyber Physical Systems-157**
- **Public Policy-157**
- **Atmospheric and Climate Sciences-157**

Branch	Engg. (E)	Professional (P) *Core + Elective*	Humanities (H)	Sciences (S) *Core+ Elective*	Un-allotted credits	ID-DD Credits	Total
AE	56	159+27	27	84 + 9	31	157	**550**
BE	45	124 + 71	27	75 + 9	42	157	**550**
BS	22	201 + 18	27	74 + 9	42	157	**550**
CH	48	155 + 45	27	75 + 9	34	157	**550**
CE	46	155 + 30	27	75 + 9	51	157	**550**
CS	45	120 + 84	27	84	33	157	**550**
ED	44	181 + 18	27	69+9	45	157	**550**
EE	48	117 + 65	27	66 + 18	52	157	**550**
ME	45	150 + 54	27	75 + 9	33	157	**550**
MM	45	165 + 27	27	66 + 18	45	157	**550**
NA	48	146 + 54	27	66 + 18	34	157	**550**
EP	45	142+63	27	75+9	32	157	**550**

Inter Disciplinary Dual Degree Programme
Category and Branch-wise credit requirements

Applicable for the following ID-DD programmes

- **Robotics-160**
- **Quantum Science and Technology (QuST)-160**
- **Complex Systems and Dynamics-160**
- **Electric Vehicles-160**
- **Quantitative Finance-160**

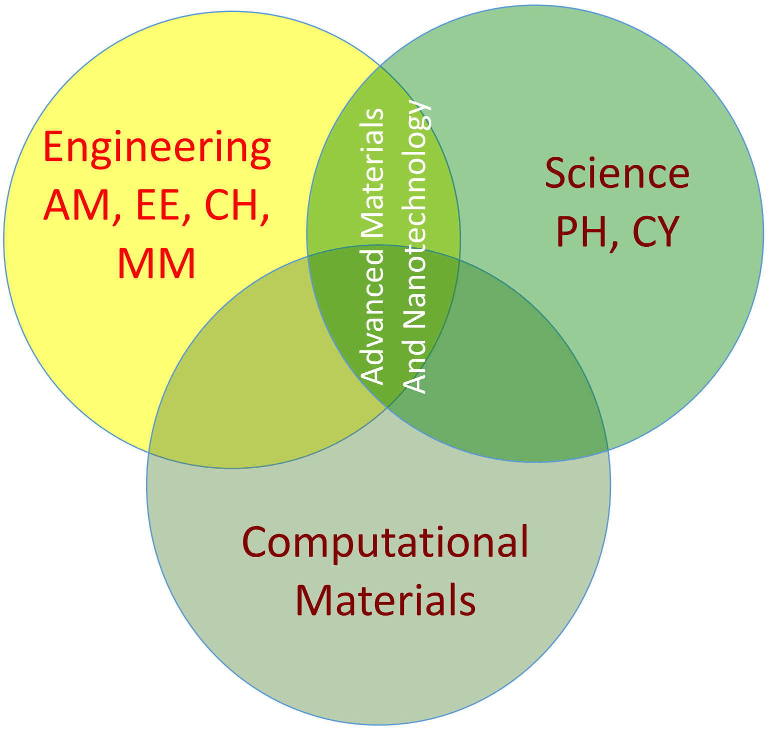
Branch	Engg. (E)	Professional (P) *Core + Elective*	Humanities (H)	Sciences (S) *Core+ Elective*	Un-allotted credits	ID-DD Credits	Total
AE	56	159 + 27	27	84 + 9	28	160	**550**
BE	45	124 + 71	27	75 + 9	39	160	**550**
BS	22	201 + 18	27	74 + 9	39	160	**550**
CH	48	155 + 45	27	75 + 9	31	160	**550**
CE	46	155 + 30	27	75 + 9	48	160	**550**
CS	45	120 + 84	27	84 + 0	30	160	**550**
ED	44	181 + 18	27	69+9	42	160	**550**
EE	48	117 + 65	27	66 +18	49	160	**550**
ME	45	177 + 27	27	75 + 9	30	160	**550**
MM	45	165 + 27	27	66 + 18	42	160	**550**
NA	48	146 + 54	27	66 + 18	31	160	**550**
EP	45	142 + 63	27	75 + 9	29	160	**550**

Inter Disciplinary Dual Degree Programme
Category and Branch-wise credit requirements for IDDD Programme - techMBA

Branch	Engineering (E)	Professional (P) *Core + Elective*	Humanities (H)	Sciences (S) *Core + Elective*	Un-allotted credits	techMBA credits	Total
AE	56	159 +27	27	84 + 9	14	**174**	**550**
BE	45	124 + 71	27	75 + 9	25	**174**	**550**
BS	22	201 + 18	27	74 + 9	25	**174**	**550**
CH	48	156 + 45	27	75 + 9	16	**174**	**550**
CE	46	155 + 30	27	75 + 9	34	**174**	**550**
CS	45	120 + 84	27	84	16	**174**	**550**
ED	44	181 + 18	27	69+9	28	**174**	**550**
EE	48	118 + 65	27	66 + 18	35	**174**	**550**
ME	45	177+27	27	75 + 9	16	**174**	**550**
MM	45	165 + 27	27	66 + 18	28	**174**	**550**
NA	48	146 + 54	27	66 + 18	17	**174**	**550**
EP	45	142 + 63	27	75+9	15	**174**	**550**

Interdisciplinary Dual Degree in Advanced Materials and Nano Technology

The world around us is made of materials of various kinds and many of these are at the heart of great technological innovations. In the recent times, with the development of nanotechnologies, the functionalities of conventional materials have advanced further and many novel applications are now being explored. Such advanced materials, both in the conventional (bulk) and nano form, are important in several fields such as energy conversion (solar cells) and storage (batteries), microelectronic devices, multiferroic materials, bio-compatible coatings and implants, high strength materials and functional materials for sensors, membranes *etc*. This interdisciplinary Dual Degree (ID-DD) program aims at equipping the students with an understanding of the fundamental science behind advanced materials and also training them with the practical tools and techniques of fabrication (materials and devices). The Department of Physics, IIT Madras is coordinating this new program with the active participation of several other departments.



The programme

The ID-DD programme in Advanced Materials and Nanotechnology designed with inputs from Engineering and Science disciplines. Departments currently offering courses towards this program are Applied Mechanics, Chemical, Chemistry, Electrical, Materials and Metallurgy, Physics. It is typically coordinated by one of these departments. Currently it is coordinated by the Department of Physics.

Eligiblity

B.Tech students (all branches) of IIT Madras can opt for this program. The minimum eligibility criteria prescribed by the senate is that the student should have a minimum of 8 CGPA at the end of 5^{th} semester.

Curriculum

The Curriculum is designed with the understanding that the modern field of Advanced materials and Nanotechnology is based on the exchange of ideas between sciences and Engineering branches. Synthesizing novel materials involves a good knowledge of chemistry and the physical properties exhibited by these novel materials are studied by the Material Scientist (cutting across Physics, Chemistry and Materials departments). Application of these materials for wider usage needs the involvement of Engineers. Computational material science forms an integral part of the field of advanced materials both in terms of the design and understanding of the physics involved. Application of these materials encompasses a wide range of fields: Energy generation and storage materials, batteries, micro electronic devices, magnetic materials, water purification, high strength materials, sustainable plastics, sensors, etc.

The curriculum is expected to facilitate both short/long term internships with private and public companies having strong research and development wings. These internships can lead to final-year projects towards the Dual Degree programme. These internships and projects are expected to enhance industry-academia collaboration for development of novel engineering products based on advanced materials and nano systems

The list of courses to be offered for students opting for DD in Advanced Materials and Nano Technology will have both core courses and electives. The students are allowed to choose four electives out of a total of 33 electives cutting across different disciplines. There will be four core courses that include a course on the Science and Technology of Solid State, two courses on nanomaterial's and nanotechnology and a Laboratory course aimed at giving hands-on experience on Advanced materials and nano systems (36 credits of DD core course).

Those students who opt for the Dual Degree program will do the courses from their 7th Semester as prescribed below.

S.No	Course No.	Course Name	L	T	E	P	O	C
Semester VII								
1	PH5011	*Core 1:* Science and Technology of Solid State	3	1	0	0	6	10
2	PH6022	*Core 2:* Introduction to Nanoscience	3	0	0	0	6	9
Total credits			**6**	**1**	**0**	**0**	**12**	**19**
Semester VIII								
1	*PH6011*	*Core 3: Nano materials and nanotechnology*	3	0	0	0	6	9
2	Elective 1	To be taken from the list of Electives mentioned	3	0	0	0	6	9
3	Elective 2	To be taken from the list of Electives mentioned	3	0	0	0	6	9
4	PH6015	*Core 4:* Advanced Materials and Nanotechnology Lab	0	0	0	6	2	8
Total Credits			**9**	**0**	**0**	**6**	**20**	**35**
Semester IX								
1	ID5190	Project-I (Summer)/Summer internship	0	0	0	0	25	25
2	Elective 3	To be taken from the list of Electives mentioned	3	0	0	0	6	9
3	Elective 4	To be taken from the list of Electives mentioned	3	0	0	0	6	9
3	ID5191	Project II (In the institute)	0	0	0	0	20	20
Total Credits			**6**	**0**	**0**	**0**	**57**	**63**
Semester X								
1	ID5192	Project III (in the Institute)	0	0	0	0	40	40
Total Credits			**0**	**0**	**0**	**0**	**40**	**40**

Total credits for the IDDD programme: 157

Core Courses								
1	PH5011	Science and Technology of Solid State	3	1	0	0	6	10
2	PH6022	Introduction to Nanoscience	3	0	0	0	6	9
3	PH6011	Nano materials and nanotechnology	3	0	0	0	6	9
4	PH6015	Advanced Materials and Nanotechnology Lab	0	0	0	6	2	8

List of Electives								
1	PH5310	Synthesis of Functional Materials	3	0	0	0	6	9
2	PH5320	Techniques of Physical measurements	3	0	0	0	6	9
3	PH5730	Methods of Computational Physics	3	0	0	0	6	9
4	PH5670	Physics and Technology of Thin Films	3	0	0	0	6	9
5	PH5690	Applied Magnetics	3	0	0	0	6	9
6	PH5600	Physics at Low Temperatures	3	0	0	0	6	9
7	PH5680	Superconductivity and applications	3	0	0	0	6	9
8	PH6013	Functional materials, Sensors and Transducers	3	0	0	0	6	9
9	PH5813	Principles of nanophotonics	3	0	0	0	6	9
10	PH5462	Magnetism in solids	3	0	0	0	6	9
11	PH6012	Semiconductor Physics and devices	3	0	0	0	6	9
12	PH5660	Non-linear Optical Processes & Devices	3	0	0	0	6	9
13	EE5347	Electronic and Photonic nanoscale devices	3	0	0	0	6	9
14	EE6500	Integrated Optoelectronics Devices and Circuits	3	1	0	0	6	10
15	ID6102	Principles and techniques of Transmission Electron Microscopy	3	0	0	0	6	9
16	ID5010	Advanced materials and processing	3	0	0	0	6	9
17	ID6050	Chemical Physics of Modern Technical Ceramics	3	0	0	0	6	9
18	MM5210	X-ray diffraction techniques	3	0	0	0	6	9
19	MM5680	Smart Materials	3	0	0	0	6	9
20	MM5700	Topics in nanomaterials	3	0	0	0	6	9
21	MM5017	Electronic materials devices and Fabrication	3	0	0	0	6	9
22	ME7023	Foundations of Computational Materials Modeling	3	0	0	0	6	9
23	CY6380	A Chemical Approach to Nanomaterials	3	0	0	0	6	9
24	CH5012	Modeling and Simulation of Particulate Processes	3	0	0	0	6	9
25	CH5021	Molecular Simulation of Soft Matter	3	0	0	0	6	9
26	CH5270	Polymers for Devices	3	0	0	0	6	9
27	CY6118	Experimental methods in chemistry	3	0	0	0	6	9
28	ID6030	Introduction to nano science and nanotechnology	3	0	0	0	6	9
29	EE5343	Solar Cell Device Physics and Materials Technology	3	0	0	0	6	9
30	EE5346	Introduction to plastic electronic	3	0	0	0	6	9
31	EE5340	MicroElectroMechanical Systems	3	0	0	0	6	9

32	EE5312	VLSI Technology	3	1	0	0	6	10
33	CH5190	Introduction to Macromolecules	3	0	0	0	6	9
34	MM5041	Medical Materials	3	0	0	0	6	9
35	MM5460	Physical Ceramics	3	0	0	0	6	9
36	AM5470	Analysis & Design of Smart Material Structure	3	0	0	0	6	9
37	AM6190	Cellular structures and mechanics	3	0	0	0	6	9
38	AM6512	Application of Molecular Dynamics	3	0	0	0	6	9

Potential recruiters

Based on the specialization the students opt for, they can be recruited by one of the private or public sector companies mentioned below. They will be perfectly suited to the research and product innovation wings of these companies.

1. Nissan
2. CUMI
3. Samsung
4. Honeywell
5. MRF
6. Saint Gobain
7. Titan
8. Aditya Birla Science and Technology Private Limited
9. IBM (R & D)
10. TATA Centre
11. SITAR, Bangalore
12. SAMEER, Chennai
13. APPLIED MATERIALS
14. DRDO, CSIR, ISRO, BHEL, DMRL, NML, NAL, ARCI, HAC

Interdisciplinary Dual Degree in Biomedical Engineering (ID-DD-BME)

The Interdisciplinary Dual Degree programme in Biomedical Engineering is intended to produce graduates with up-to-date and fundamental understanding of biomedical engineering, by integrating various engineering disciplines with biomedical sciences. This programme aims to produce graduates who are ready to hit the ground running in industry, as well as foster new knowledge and evolve leadership in biomedical engineering research and entrepreneurship.

Who offers the programme?

The ID-DD programme is championed by the Dept of Applied Mechanics together with Depts. of Biotechnology, Electrical Engineering, Computer Science and Engineering, Mechanical Engineering, Materials and Metallurgical Engineering, Physics and Mathematics. The true interdisciplinary nature of Biomedical Engineering is reflected in the joint programme collectively offered by various allied Depts.

Who can enrol in this programme?

A B. Tech student of IIT Madras in any discipline is eligible to upgrade to this programme provided the students meets certain minimum academic norms. Selection of applicants will be based on academic performance and an interview to ascertain aptitude.

What is the curriculum?

The students graduating under this programme are trained in fundamental engineering sciences as well as application oriented skills in specific areas of biomedical engineering. This programme includes an optional introductory section in the form of elective modules for the entire undergraduate community of IIT Madras. These modules enable students to appreciate the exciting possibilities in Biomedical Engineering and explore the option of further studies in the form of a dual Masters' degree by enrolling in the programme.

ID-DD-BME has a very flexible curriculum. The programme spans a period of the last two years of a five-year dual degree programme. Five out of seven theory courses are flexible, in order to enable the students to choose from four major streams of biomedical engineering such as biomaterials, bioinstrumentation, image and signal processing and medical physics. This curriculum enables the students to explore diverse fields of biomedical engineering matched to the skills acquired as part of their UG curriculum. The interdisciplinary lab sessions in the curriculum range from circuit building exercises to cutting edge research-oriented experiments at various participating laboratories.

Frequent field visits to reputed medical educational institutions, research centres as well as hospitals help the students to understand the current clinical needs to evolve ideas for product oriented research. These students will emerge as an excellent fit for application-oriented product development. The curriculum also allows short term (1-3 months)/ long term (up to 6 months) internships with potential companies / research organizations which could be extended to a full-year project to meet part of their credit requirements, wherever the project is deemed to fit our academic standards. Such projects could enable industrial collaboration for the development of indigenous products in the healthcare industry.

Interdisciplinary DD in Biomedical Engineering-course curriculum

	Semester 7							
1	AM5119	Core 1: Physiology for Engineers	3	0	0	0	6	9
2	AM5010	Core 2: Biomechanics	3	0	0	0	6	9
3	Elective 1	Elective 1: To be selected from BME / core basket	3	0	0	0	6	9
4	AM5023	Physiological measurements and Instrumentation Laboratory	0	0	0	2	2	4
		Total credits						**31**

	Semester 8							
1	MM5040	Core 3: Medical materials	3	0	0	0	6	9
2	AM5XXX	Core 4: To be selected from basket of core courses	3	0	0	0	6	9
3	Elective 2	Elective 2: To be selected from BME / core basket	3	0	0	0	6	9
4	AM5019	Advanced BME lab	0	0	0	3	2	5
		Total Credits :						**32**

	Semester 9							
1	ID5290	Project-I (Summer) / Summer internship	**0**	**0**	**0**	**0**	**15**	**15**
2	Elective3	Elective 3: To be selected from BME / core basket	3	0	0	0	6	9
3	ID5291	Project II	0	0	0	0	30	30
		Total Credits :	**3**	**0**	**0**	**0**	**54**	**54**

	Semester 10							
1	ID5292	Project III	0	0	0	0	40	40
		Total Credits :						**40**

Total credits for the DD programme: 157

		Basket of core courses	**L**	**T**	**+**	**P**	**O**	**C**
1	AM 5160	Biomedical Imaging systems	3	0	0	0	6	9
2	AM5130	Quantitative physiology	3	0	0	0	6	9
3	AM5140	Biomedical instrumentation	3	0	0	0	6	9
4	AM5510	Biomedical Signals and Systems	3	0	0	0	6	9
5	AM 5050	Biomedical sensors and measurements	3	0	0	0	6	9

		Elective streams (Basket of Biomedical Electives)						
	Stream 1	**Biomechanics**						
1	PH5730	Methods of Computational Physics	3	0	0	0	6	9
2	ME6000	Computational Methods in Engg	3	0	0	0	6	9
3	MA6270	Numerical solutions of partial differential equations	3	0	0	0	6	9
4	MA5890	Numerical linear algebra	3	0	0	0	6	9
5	AM 7010	Classics in Neuroscience	3	0	0	0	6	9
6	AM 5170	Orthopedics Mechanics	3	0	0	0	6	9
7	AM 6190	Movement disorders and neurorehabilitation	3	0	0	0	6	9
8	AM 5190	Haptics and biomedical engineering	3	0	0	0	6	9
9	AM 5060	Psycophysics	3	0	0	0	6	9
10	AM 6516	Neuromechanics of human movement	3	0	0	0	6	9
11	ME6012	Mechanics of human movement	3	0	0	0	6	9
12	AM 5110	Biofluid mechanics	3	0	0	0	6	9
	Stream 2	**Biomedical instrumentation**						
1	AM 5050	Biomedical sensors and measurements	3	0	0	0	6	9
2	AM 5140	Biomedical instrumentation	3	0	0	0	6	9
3	AM 5100	Biomedical laser instrumentation	3	0	0	0	6	9
4	AM5160	Biomedical imaging systems	3	0	0	0	6	9
5	AM 5013	Operating theatre instrumentation and surgical tech technology	3	0	0	0	6	9
6	AM 5115	Systems approach in Biomedical engineering	3	0	0	0	6	9
10	EE6403	Transducers for instrumentation	3	0	0	0	6	9
11	EE6402	Biomedical electronic systems	3	0	0	0	6	9
12	EE6501	Optical sensors	3	1	0	0	8	12
13	EE5502	Optical engineering	2	3	0	0	7	12
	Stream 3	**Medical Physics**						
1	AM 6518	Biophysical aspects of tumor microenvironment	3	0	0	0	6	9
2	AM 5190	Cellular structures and mechanics	3	0	0	0	6	9
3	AM 5120	Biomaterials	3	0	0	0	6	9
4	BT5011	Biomaterials engineering	3	0	0	0	6	9
5	EE5500	Introduction to photonics	3	0	0	0	6	9
6	EE6506	Computational electromagnetics	4	0	0	0	0	12
	Stream 4	**Image and Signal processing in Biomedical Engineering**						
1	AM4010	Biomedical signal processing	3	0	0	0	6	9
2	CS6300	Speech Technology	3	0	0	0	6	9
3	CS6690	Pattern recognition	3	0	0	0	6	9
4	AM5020	Biomedical Ultrasonics	3	0	0	0	6	9
5	EE4240	Image signal processing	3	0	0	0	6	9

Who are the potential recruiters?

There are several leading companies who actively recruit students with knowledge of biomedical engineering. Some of the companies are:

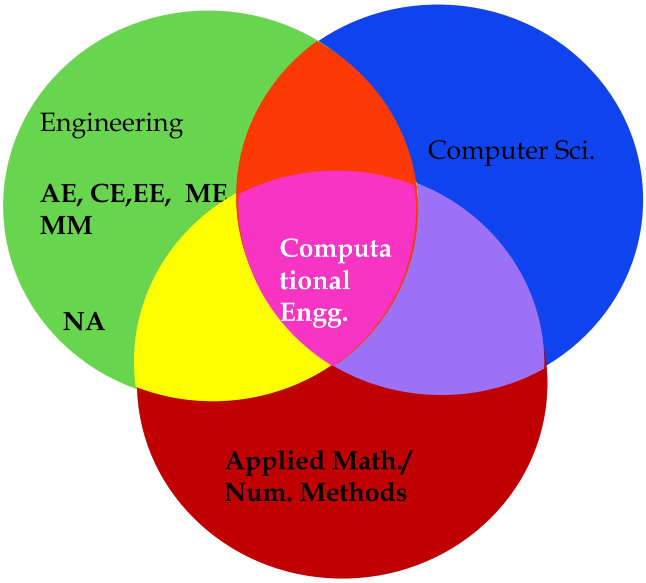
• Accenture Solutions Pvt Ltd
• American Express
• Analog Devices
• Biocon Ltd.
• Cerner
• Ernst and Young
• Flutura Business Solutions Pvt. Ltd.
• GE Healthcare
• Grind Master Machines Pvt. Ltd.
• J Mitra
• Johnson and Johnson
• Medtronic
• Olympus medical systems India Pvt Ltd.
• Philips India Ltd (Philips healthcare)
• Schlumberger
• Siemens Technology and Services Pvt. Ltd.
• Stryker
• TCS Labs

There are several renowned universities which have strong biomedical engineering research programmes, some of whom collaborate with IIT Madras. Some of them are:

- Queensland University of Technology
- RWTH Aachen

Interdisciplinary Dual Degree in Computational Engineering (ID-DD-CE)

The development of Engineering Analysis and design tools for solving Complex Engineering problems is facilitated in the Interdisciplinary Dual Degree programme in Computational Engineering. Computing tools for Engineering software that involve CPU intensive calculations as their backbone are pervasive in disciplines such as, Aerospace, Civil, Chemical, Electrical, Mechanical, Materials, Naval Engineering etc. The graduates from this ID program will strengthen their Simulation and Mathematical modelling expertise in the core Engineering discipline by learning relevant tools and techniques from Computer Science and Applied Mathematics in a structured and systematic way.



Who offers the programme?

The ID-DD programme in Computational Engineering is drawn from a wide spectrum of disciplines in Engineering and Science. Some of the Departments who are offering these courses include, Mechanical, Materials & Metallurgy, Mathematics, Physics, Ocean, Applied Mechanics, Civil, Chemical, etc. For the purpose of administrative ease, it is typically governed by one of these participating departments.

Who can enrol in this programme?

A B. Tech student of IIT Madras from any discipline is eligible to upgrade to this programme provided the student meets certain minimum academic norms. Selection of applicants will be based on the CGPA cut off.

What is the curriculum?

The core philosophy of the curriculum is that, complex Engineering problems do not have simple closed form solutions. Hence, advanced numerical methods (such as, Finite Volume and Finite Element Methods) need to be used for solving these problems. The analysis and design of such systems necessitate solving a system of linear/ non-linear simultaneous equations running into millions/ billions of degrees of freedom. Hence, the computational skill set of understanding algorithms and deployment of suitable data structures to implement them into useful codes is necessary. Furthermore, tools and techniques from high performance computing will facilitate efficient computation and code parallelization. Similarly skill set from discretization methods Engineering Mathematics will be imparted.

The IDDD-CE curriculum facilitates short term (1-3 months)/ long term (up to 6 months) internships with potential companies / research organizations which could be extended to a full-year project to meet part of their credit requirements, wherever the project is deemed to fit our academic standards. Such projects would promote industry-academia collaboration for the development of practical engineering solutions. .

The curriculum for the IDD - Computational Engineering

Sl. No	Course No	Course Name	L	T	E	P	O	C
	Semester7							
1	Core - 1	**CORE-1 basket**	3	0	0	0	6	9
2	Core - 2	**CORE-2 basket**	3	0	0	0	6	9
3	Elective - 1	Elective 1: Preferably chosen from a chosen elective stream	3	0	0	0	6	9
4	AM5801	Computational Laboratory	0	0	0	2	2	4
		Total credits						**31**

	Semester 8							
1	Core – 3	**CORE - 3 basket**	3	0	0	0	6	9
2	AM5080	**CORE – 4** High Performance Computing for Engineering Applications	3	0	0	3	8	14
3	Elective- 2	Elective 2: Preferably chosen from the same Elective stream	3	0	0	0	6	9
		Total Credits :						**32**

	Semester 9							
1	ID5390	Project-I (Summer) / Summer internship	**0**	**0**	**0**	**0**	**15**	**15**
2	Elective - 3	Elective 3: Preferably chosen from the same Elective Stream	3	0	0	0	6	9
3	ID5391	Project II	0	0	0	0	30	30
		Total Credits :	**3**	**0**	**0**	**0**	**54**	**54**

	Semester 10							
1	ID5392	Project III	0	0	0	0	40	40
		Total Credits :						**40**

- **Total credits for the DD program : 157 (121+36)**
- #, *Project grade assigned at the end of10th semester
- Electives / cores from below list or any relevant courses from other Departments could be chosen in consultation with Faculty Advisor.

Basket of courses for CORE - 1 : Numerical Methods

1	AM5600	Computational Techniques in Applied Mechanics	3	0	0	0	6
2	ME6000	Computational Methods in Engineering	3	0	0	0	6
3	ME6150	Numerical Methods in Thermal Engineering	3	0	0	6	6
4	MA5470	Numerical Analysis	3	0	0	0	6
5	PH5730	Methods of Computational Physics	3	0	0	0	6
6	CH6060	Numerical Techniques for Engineers	3	0	0	0	6
7	MM5024	Numerical Methods for Metallurgists	3	0	0	0	6
8	OE5450	Numerical Techniques in Ocean Hydrodynamics	3	0	1	0	6

Basket of courses for CORE - 2 : Computational Implementation

1	MA5901	Data Structures and Algorithms	3	0	0	0	6
2	ID6105	Computational Tools: Algorithms, Data Structures and Programs	3	0	0	0	6

Basket of courses for CORE - 3: Discretization Methods

1	CE5610	Finite Element Analysis	3	0	0	0	6
2	AM5450	Fundamentals of Finite Element Analysis	3	0	0	0	6
3	CH6110	Finite Element Methods in Engg	3	0	0	0	6
4	ME6800	Finite Element Analysis	3	0	0	0	6
5	AM5630	Foundations of Computational Fluid Dynamics	3	0	0	0	6
6	OE5500	FEM applied to Ocean Engineering	3	0	0	0	6
7	CH6020	Computational Fluid Dynamics Techniques	3	0	0	0	6
		Elective streams					

	Stream 1	**Computational Fluid Dynamics**					
1	AM5630	Foundations of Computational Fluid Dynamics	3	0	0	0	6
2	AM5570	Introduction to Turbulence	3	0	0	0	6
3	AM6513	Advanced Computational Fluid Dynamics	3	0	0	0	6
4	AM 5640	Turbulence Modeling	3	0	0	0	6
5	ME6650	Computational Fluid Dynamics of Turbomachinery	3	0	0	0	6
6	ME6006	Computational Heat and Fluid Flow	3	0	0	0	6
7	CH6020	Computational Fluid Dynamics Techniques	3	0	0	0	6
8	AM6512	Application of Molecular Dynamics	3	0	0	0	6
9	ME6280	Design and Optimization of Energy systems	3	0	0	0	6
10	OE6020	Meshfree methods applied to hydrodynamics	3	0	3	0	6
11	PE6031	Reservoir Simulation	3	0	0	0	6
12	AM5530	Advanced Fluid Mechanics	3	0	0	0	6
13	CH 5140	Process Analysis and Simulation	3	0	0	0	6
14	CH 5541	Advanced Momentum Transport	3	0	0	0	6
	Stream 2	**Computational Solid Mechanics**					
1	AM5450	Fundamentals of Finite Element Analysis	3	0	0	0	6
2	AM6512	Application of Molecular Dynamics	3	0	0	0	6
3	AM6291	Computational Structural Dynamics	3	0	0	0	6
4	ME7680	Optimization Methods for Mechanical Design	3	0	0	0	6
5	ME6280	Design and Optimization of Energy systems	3	0	0	0	6
6	CE7730	Advanced Finite Element Analysis	3	0	0	0	6
7	AM5390	Advanced Structural Mechanics	3	0	0	0	6
	Stream 3	**Computational Materials Engineering**					
1	ME7023	Foundations of Computational Materials Modeling	3	0	0	0	6
2	MM6010	Computational Materials Thermodynamics	3	0	0	0	6
3	ME7160	Computational Methods in Design & Mfg.	3	0	0	0	6
4	AM6512	Application of Molecular Dynamics	3	0	0	0	6
5	MM5011	Modeling of Transport Phenomena in multi-phase systems	3	0	0	0	6
6	MM5003	Atomistic Modeling of Materials	2	1	0	0	6
	Stream 4	**Computational Biology**					
1	BT6090	Intro. to Bioinformatics & Computational Biology	3	0	0	0	6
2	BT6270	Computational Neuroscience	3	0	0	0	6
3	BT5420	Computer Simulations of Biomolecular Systems	3	0	0	0	6
4	BT5240	Computational Systems Biology	3	0	0	0	6
5	ME5560	Heat and Mass Transfer in Biological Systems	3	0	0	0	6

Who are the potential recruiters?

Depending the path the students build for themselves, they are expected to be placed in most of the private and public sector companies mentioned below. They will be mostly working in the R&D group and/or Analysis and Design groups of these industries.

(i) GE Air-craft engines, GE health care

(ii) General Motors, Tata Motors, TVS, Mahindra & Mahindra,

(iii) Forbes-Marshal, FL Smidt, Reliance, Thermax, Alpha-Laval,

(iv) Tata Steel, Bao Steel, Mittal Steel, Coal India

(v) Fluent, ANSYS, ABACUS, Numeric, ,

(vi) TRDDC, TERI, NPCIL, NTPC, GTRE, CVRDE, BDL, BHEL, BEL, CDAC

(vii) ONGC, OIL, HP, BP, GAIL

(viii) Engineering divisions of Software companies: TCS, Infosys, Wipro, Hexaware,

There are several renowned universities which offer strong Computational engineering Courses and research based programmes, some of them are:

- UIUC
- MIT
- Georgia Tech
- Texas A & M
- NUS

Interdisciplinary Dual Degree in Data Science
(IDDD-DS)

With the tremendous availability of large volumes of data across several domains there has been an explosion of interest in all aspects of handling and understanding data. Data Science brings together all aspects of technology required for gathering, storing, analyzing and understanding data. This includes storage technology, distributed computing, data driven modeling, data analytics and mining, visualization, security, etc. Given that proper interpretation and modeling requires good domain understanding this becomes inherently an interdisciplinary endeavour.

Who offers the programme?

The goal of this program is to give basic background to students from different disciplines in data science and provide ample opportunity for them to specialize in a particular aspect of data science through the electives and the project. Given that often data science is associated with computing, and given that advances in computing technology have enabled the whole field, Computer Science Engineering (CSE) is the natural department to host this program. The program will have a strong interdisciplinary flavor with many departments participating in the teaching of the courses and guidance of the students.

Who can enrol in this programme?

A B. Tech student of IIT Madras from any discipline is eligible to upgrade to this programme provided the student meets certain minimum academic norms. Selection of applicants will be based on the CGPA cut-off of 8.0 at the end of the 5th semester.

What is the curriculum?

The curriculum has a core component that covers the fundamental theoretical concepts and tools required. The student is then free to choose 3-4 electives from the prescribed list. These electives are a mix of advanced algorithmic or theoretical courses and applied data science courses. Depending on the interests of the students one can choose to specialize in a specific application area or acquire deeper grounding the fundamentals of data science.

THE CURRICULUM FOR THE IDDD – DATA SCIENCE

6th Semester

S.No	Course No.	Course Name	L	T	E	P	O	C
1	CH5019	Mathematical Foundations for Data Science	4	0	0	0	8	12
		Total Credits :						**24**

7th Semester

S.No	Course No.	Course Name	L	T	E	P	O	C
1	EE5708	Data Analytics Laboratory Offered by Department of Electrical Engg. Covers basics of python or R. Simple analytics tasks - regression, classification, clustering, associations, etc. Emphasis will be on choice of models, evaluation of results, significance analysis, visualization and interpretation of results.	0	0	0	3	3	6
2	ID5055	Foundations of Machine Learning	4	0	0	0	8	12
3	Elective 1							9
		Total Credits :						**27**

8th Semester

S.No	Course No.	Course Name	L	T	E	P	O	C
1	CS5830	Big Data Laboratory Offered by Department of Computer Science and Engineering Will cover basics of Map-Reduce, Distributed data storage, Spark/Hadoop; Working on the cloud - Amazon Web Services or Azure as a case study; Emphasis would be on data analytics use cases.	0	0	0	3	3	6
2	Elective 2	To be chosen from a basket of Data Science						9
3	Elective 3	To be chosen from a basket of Data Science						9
4	Elective 4	To be chosen from a basket of Data Science						9
		Total Credits :						**33**

SUMMER

S.No	Course No.	Course Name	L	T	E	P	O	C
1	ID5490	Project I	0	0	0	0	20	20
		Total Credits :						**20**

9th Semester

S.No	Course No.	Course Name	L	T	E	P	O	C
1	ID5491	Project II	0	0	0	0	30	30
		Total Credits :						**30**

10th Semester

S.No	Course No.	Course Name	L	T	E	P	O	C
1	ID5492	Project III	0	0	0	0	35	35
		Total Credits :						**35**

Electives: 36 credits from the approved list of electives. **Suggested Electives (Will be updated when newer electives are offered)**

Total credits for the DD program: 157 (121+36)

Odd Semester

Course No.	Course Title	Faculty
CH5170	Process Optimization	Sridharakumar Narasimhan
CHXXXX	AI in Process Engineering	Raghunathan Rengasamy
CH5350	Applied Time-Series Analysis	Arun K. Tangirala
CH5020	Statistical Analysis and Design of Experiments	Arun K. Tangirala / A. Kannan
CS5011	Introduction to Machine Learning	B. Ravindran
CS6370	Natural Language Processing	Sutanu Chakraborti
CS6740	Searching and Indexing in Large Datasets	Sayan Ranu
CS6310	Artificial Neural Networks	Anurag Mittal
MA5750	Applied Statistics	Neelesh Upadhye
MA5014	Applied Stochastic Processes	Neelesh Upadhye
MA5013	Applied Regression Analysis	Neelesh Upadhye
EE5177	Machine Learning for Computer Vision	Kaushik Mitra
CE7011	Advanced Transportation Network Analysis	Karthik Srinivasan/ Gitakrishnan Ramadurai

Even Semester

Course No.	Course Title	Faculty
BT5240	Systems Biology	Karthik Raman
CH5440	Multivariate Data Analysis	Shankar Narasimhan
CH5230	System Identification	Arun K. Tangirala
CH5470	Graph Theory & Its Applications in Process Design	Sridharakumar Narasimhan
CS6720	Data Mining	Sayan Ranu
CS6700	Reinforcement Learning	B. Ravindran
CS6011	Kernel Methods for Pattern Analysis	Chandrasekhar C.
CS6770	Knowledge Representation & Reasoning	Deepak Khemani
CS6730	Probabilistic Reasoning in AI	B. Ravindran
CS6012	Social Network Analysis	B. Ravindran
EE5154	Complex Network Analysis	Venkatesh R
MS6032	Predictive and Prescriptive Data Analytics	Nandan Sudarsanam
CS6741	Algorithms for Big Data	John Augustine
CHXXXX	Manufacturing Analytics	Raghunathan Rengasamy
CS7015	Deep Learning	Mitesh Khapra
CE5390	Analytical Techniques in Transportation Engineering	Karthik Srinivasan
CE5290	Transportation Network Analysis	Karthik Srinivasan

Interdisciplinary Dual Degree in Energy Systems (ID-DD-ES)

The Interdisciplinary Dual Degree Programme in Energy Systems is intended to equip the student with the necessary skills to deal with the fast evolving energy related technologies of our time. The need for an interdisciplinary approach in the energy domain is increasingly felt, since the technology of energy conversion should necessarily involve considerations of usage, materials needed and the commercial and environmental aspects of the processes. Towards this end, this programme enables the student to understand the various dimensions of energy usage and conversion and makes them ready to tackle the complex realities that exist in the field.

Who Offers the Programme?

Many Departments in IITM have come together to offer this programme. The Departments of Aeronautical, Applied Mechanics, Chemical, Civil, Electrical, Mechanical, Metallurgical and Materials Engineering, Ocean Engineering, along with the Departments of Chemistry, Physics, Humanities and Social Sciences, have all come together to enable a grasp of the multidimensional aspect of energy, technology and society.

Who is Eligible to take the Programme?

Any BTech student of the Institute with certain minimum norms of academic performance can apply for this programme in their fifth semester.

What is the Curriculum?

The curriculum consists of eight courses. These are done in the seventh to tenth semesters of the dual degree programme. There are four core courses which will be taken by all entrants: Principles of Thermal Energy Conversion, Renewable Energy Technology, Materials for Energy Storage and Conversion, and Energy Economics. These subjects enable the student to get an exposure to the vast and multidimensional impact that the energy domain has.

The courses on Thermal Energy Conversion and Materials are to be done in the seventh semester. The former enables an understanding of various gas and vapour cycles and focusses for a large part on Thermal power plants and coal combustion. Nuclear Reactor principles are also included in this. The latter course enables the student to get a good understanding of materials and technologies for material synthesis. Materials for batteries, fuel cell technologies and supercapacitors are also part of this course.

The course on Renewable Energy Technology deals with various types of renewable energy sources and their usage, ranging from solar, wind to geothermal and bio-fuels. The Energy Economics course, which is again in the eighth semester deals with pricing, taxation, energy markets, economics of various types of sources, climate change and policy aspects.

While the core courses are designed to give a complete overview of the entire domain, the students are free to choose electives that will enable them to chalk out a further path of their choice. A large set of carefully selected electives are provided which will enable the student to explore a particular aspect of the energy domain in greater detail. The curriculum also stipulates that one elective be done from a basket of courses dealing with the final utilization of energy.

As an example, a student interested in wind power may choose the following electives: Wind Turbines, Power Electronics, Power Quality and Distributed Generation. On the other hand, doing all electives in Stream C (Wind and Ocean Energy Systems) together with Powering and Propulsion of Marine Vehicles would prepare the student for more details of possible energy activities off-shore. A set of electives such as Principles of Fuel Cells, Chemical and Electrochemical Energy Systems, Power Electronics and

Intelligent Transportation may ready a student more towards the automotive uses of energy. It is also of course possible to take electives from a specific stream as listed (see below in List of Courses) to take advantage of a homogenous and planned set of courses. The possibilities are many. The courses may require pre-requisites which the student has to plan ahead and take through the free credits that are available in the overall programme. Two of these electives are to be taken in the seventh semester and two in the eighth semester.

The ninth and tenth semesters of the programme are devoted to doing a project in an area of energy.

No.	**Course No**	**Course name**	**L**	**T**	**E**	**P**	**O**	**C**
	Semester VII							
1	ME5129	Principles of Thermal Energy Conversion	3	0	0	0	6	9
2		Elective	3	0	0	0	6	9
3		Elective	3	0	0	0	6	9
4		Elective	3	0	0	0	6	9
		Total credits	12	0	0	0	24	**36**
	Semester VIII							
1	ME6148	Renewable Energy Technology	3	0	0	0	6	9
2	ID5070	Energy Economics	3	0	0	0	6	9
3	Core	**Core Basket:** 1. ID6106-Materials for Energy Storage and Conversion (9 cr) 2. MM5030- Materials in renewable energy technologies (9 cr) 3. ID6050-Chemical Physics of Modern Technical Ceramics (12 cr)						9 or 12
4		Elective	3	0	0	0	6	9
		Total Credits :	9	0	0	0	18	**36 or 39**
	Semester VIII							
1	ID5590	Project I (summer)	0	0	0	0	15	**15**
	Semester IX							
2	ID5591	Project II (during semester)	0	0	0	0	30	**30**
		Total Credits :						
	Semester X							
1	ID5592	Project III	0	0	0	0	40	**40**
		Total Credits :						**157 or 160**

		CORE COURSES						
	New	Principles of Thermal Energy Conversion	3	0	0	0	6	9
	ME6148	Renewable Energy Technology	3	0	0	0	6	9
	New	Materials for Energy Storage and Conversion	3	0	0	0	6	9
	New	Energy Economics	3	0	0	0	6	9
	Stream A	**Energy Storage Systems**						
	CH5013	Principles of Fuel Cells	3	0	0	0	6	9
	CY6114	Chemical and Electrochemical Energy Systems	3	0	0	0	6	9
	CH5022	Solar Photoelectrochemistry	3	0	0	0	6	9
	Stream B	**Materials**						
	PH 6013	Functional Materials, Sensors and Transducers	3	0	0	0	6	9
	ID6050	Chemical Physics of Modern Technical Ceramics	3	0	0	0	6	9
	MM3180	Advanced Materials and Processes	3	0	0	0	6	9
	MM 5030	Materials in Renewable Energy Technologies	3	0	0	0	6	9
	MM5460	Physical Ceramics	3	0	0	0	6	9
	Stream C	**Solar Energy Systems**						
	ME6005	Solar Energy for Process Heat & Power Generation	3	0	0	0	6	9
	EE5343	Solar Cell Device Physics and Materials Technology	3	0	0	0	6	9
	ME6580	Utilization of Solar Energy	3	0	0	0	6	9
	Stream D	**Wind and Ocean Energy Systems**						
	AS5450	Wind Turbines	3	0	0	0	6	9
	OE4340	Ocean Energy Systems	3	0	0	0	6	9
	OE5030	Wave Hydrodynamics	3	0	0	0	6	9
	Stream E	**Combustion**						
	ME6110	Combustion Technology	3	0	0	0	6	9
	ME6020	IC Engine Combustion and Pollution	3	0	0	0	6	9
	ME6440	Alternative Fuels for IC Engines	3	0	0	0	6	9
	Stream F	**Thermal Energy**						
	ME6570	Thermal Energy Conservation	3	0	0	0	6	9
	ME6004	Micro and Nanoscale Energy Transport	3	0	0	0	6	9
	ME6030	Refrigeration and Cryogenics	3	0	0	0	6	9
	ME5134	Process Simulation	3	0	0	0	6	9
	ME6280	Design and Optimization of Energy Systems	3	0	0	0	6	9
	New	Geothermal Energy	3	0	0	0	6	9
	Stream G	**Fuels**						
	AM5114 / PE6030	Flow Through Porous Media / Reservoir Engineering	3	0	0	0	6	9
	PE6320	Subsea Engg for Oil and Gas fields	3	0	0	0	6	9
	PE6060	Off shore Oil and Gas Production Systems	3	0	0	0	6	9
	PE6080	Petroleum Refining Technology	3	0	0	0	6	9
	PE6312	Enhanced Oil Recovery	3	0	0	0	6	9
	CH5018	Biomass Conversion Processes and Analysis	3	0	0	0	6	9
	Stream H	**Electrical Power**						

	EE3203	Power Electronics	3	0	0	0	6	9
	EE5257	Energy Management Systems and SCADA	3	0	0	0	6	9
	EE5260	Power Quality	3	0	0	0	6	9
	Stream I	**Energy utilization**						
	ME6530	HVAC Systems and Applications	3	0	0	0	6	9
	CE6011	Smart Buildings and Automation	3	0	0	0	6	9
	CE5900	Intelligent Transportation Systems	3	0	0	0	6	9
	OE6310	Powering and Propulsion of Marine Vehicles	3	0	0	0	6	9
	EE6261	Restructured Power Systems	3	0	0	0	6	9
	EE5262	Distributed Generation and Microgrid Systems	3	0	0	0	6	9
	EE5204	Electric Vehicles and Renewable Energy	3	0	0	0	6	9

The potential recruiters from such a programme are

• ABB	• KIE Solatherm
• Amat Engineering	• Larsen and Toubro
• Abener	• NPCIL
• Abengoa	• NTPC
• Altius Consulting	• ONGC
• Applied Materials	• Powergrid
• Ather	• Reliance Industries
• Bharat Petroleum	• Schlumberger
• Bosch	• Shakti Sustainable Energy Foundation
• Cummins	• Siemens
• EESL	• Suzlon
• Forbes Marshall	• Tata BP Solar
• GE	• Tata Motors
• ICICI Lombard	• Thermax
• Indianoil	• TCE

Several renowned universities offer research programmes in energy, some of whom collaborate with IIT Madras. A few of the Universities are:

- Cornell University, Ithaca, New York, USA
- Georgia Institute of Technology, Georgia, USA
- Iowa State University, Ames, Iowa, USA
- North Carolina State University, Raleigh, USA
- RWTH Aachen, Aachen, Germany
- Texas A & M University, College Station, Texas, USA
- TU Munich, Munich, Germany
- University of Colorado at Boulder, USA
- University of Melbourne, Australia
- University of Queensland, Australia
- University of Western Australia

Interdisciplinary Dual Degree in Robotics
(ID-DD-Robotics)

The Interdisciplinary Dual Degree programme in Robotics is proposed to nurture and develop the next-generation professionals in the area of robotics who can contribute in the design, development, and implementation of robotic systems in the industry and help the industry to improve their productivity, leading to the overall economic growth of the country. IIT Madras has faculty working in the area of robotics spread across various departments. Since no single department has the critical mass to offer a dual degree program in Robotics, an interdisciplinary dual degree program is proposed. The dual degree program in Robotics will be having its focus on Design, Analysis, and Application development (new system development) and the curriculum has been developed with this focus.

Learning Outcomes:

Students graduating with a dual degree in Robotics shall be capable of understanding and analyzing the following:

1. Basic robotic technologies used across various applications
2. Kinematics, dynamics, and control of Industrial and field/service robots
3. Sensing, perception, planning, and control applied to autonomous robots
4. Application of Artificial Intelligence, Neural Networks and Reinforcement learning in Robotics
5. Hardware systems and controllers used in robotics
6. Design of robotic systems for new applications

Who offers the programme?

The ID-DD programme is offered by faculty from the departments of Aerospace Engineering, Applied Mechanics, Civil Engineering, Computer Science and Engineering, Electrical Engineering, Engineering Design, Mechanical Engineering and Ocean Engineering. The true interdisciplinary nature of Robotics is reflected in the joint programme collectively offered by faculty from various Depts.

Who can enrol in this programme?

A B. Tech student or a Dual Degree student of IIT Madras in any discipline (except biosciences) is eligible to upgrade/opt for this programme provided the student has a CGPA of 8.0 or above up to 5th semester. Total number of seats will be fixed at 25 and allocation of dual degree specialization and award of the degree will be governed by the rules of the Institute.

What is the curriculum?

ID-DD-Robotics has a very flexible curriculum. The programme spans a period of five semesters of the five-year dual degree programme. There will be a bridge course covering the basics of electrical, mechanical, and computer science fundamentals applicable to robotics. This course will ensure that the students who enters into this specialisation from different streams have the basic understanding of robotics. The curriculum also allows short term (1-3 months)/ long term (up to 6 months) internships with potential companies / research organizations.

In tune with the overall structure of the dual degree program being offered in the Institute, the number of courses to be offered and the credit distribution are as follows:

Total Credits required:	**155 to 160**
No. of PMT CORE courses to be offered:	3 (33 credits)
No. of electives to be offered:	4 (36±2 credits)
No. of labs. to be offered:	1 (6 credits)
Project work/internship	1 (85 credits)

Total credits for the ID-DD specialization: 160

Interdisciplinary DD in Robotics -course curriculum

Sl. No	Course No	Course Name	L	T	E	P	O	C
	Semester 6							
1	ID6040	Core 1: Introduction to Robotics	4	0	0	0	8	12
		Total credits						**12**
	Semester 7							
1	ED5260	Core 2: Mechanics and Control of Manipulators	4	0	0	0	8	12
		Electives						
		Total Credits :						**12**
	Semester 8							
	ID5690	**Internship/Summer Project (Project I)**						
		Electives						
	Semester 9							
1	ED5315	Core 3: Field and Service Robotics	3	0	0	0	6	9
2	IDXXXX	Core Lab1: Robotics Laboratory	0	0	0	3	3	6
	ID5691	Project II						
		Total Credits :						**15**
	Semester 10							
1	ID5692	Project III						
		Total Credits :						

Project: 85 credits to be completed in 8^{th}, 9^{th} and 10^{th} semester
Electives: 36±2 credits to be completed from the approved list in 7^{th}, 8^{th}, and 9^{th} semester

Total credits for the DD programme: 160

ELECTIVE COURSES

Electives will be offered in three baskets. Students need to choose the electives from at least two baskets (no student will be allowed to choose all the electives from one basket). Faculty/Dept. consent has been received for all the electives.

		Basket 1						
1	AS5012	Dynamics and control of rotorcraft	3	0	0	0	6	9
2	AS5040	Flight Mechanics	4	0	0	0	8	12
3	AS 5010	Aerodynamics and Aircraft Performance	3	0	0	0	6	9
4	AS5340	Advanced flight mechanics	3	0	0	0	6	9
5	AM5010	Biomechanics	3	0	0	0	6	9
6	AM5190	Haptics in Biomedical Engg	3	0	0	0	6	9
7	AM5011	Virtual Reality Engg.	3	0	0	0	6	9
8	ED5314	Design, analysis and control of Robot	3	0	0	0	6	9
9	OE 5011	Marine Robotics	3	0	0	0	6	9
10	ME7010	Microprocessor in automation	3	0	0	0	6	9
11	CE6011	Smart buildings and automation	3	0	0	0	6	9
12	ED5040	Human Anatomy Physiology and Biomechanics	4	0	0	0	8	12
13	ED5160	Automotive systems	4	0	0	0	8	12
14	AS5545	Dynamics and control of spacecraft	3	0	0	0	6	9

		Basket 1						
15	AS5570	Principles of Guidance for Autonomous vehicles	3	0	0	0	6	9
		Basket 2						
1	CS5011/ EE5177	Machine Learning for Computer Vision	4	0	0	0	8	12
2	CS6380	Artificial intelligence	4	0	0	0	8	12
3	CS6700	Reinforcement learning	4	0	0	0	8	12
4	CS7015	Deep Learning	4	0	0	0	8	12
5	CS6350/ EE5175	Computer Vision/ Image Signal Processing	4	0	0	0	8	12
6	CS6777	Optimisation for computer vision applications	4	0	0	0	8	12
7	CS5691	Pattern recognition and Machine Learning	4	0	3	0	8	15
8	CS6910	Fundamentals of Deep Learning	3	1	0	0	8	12
		Basket 3						
1	EE5541	Synthesis of control systems	3	0	0	0	6	9
2	EE6417	Allied topics in control systems	3	0	0	0	6	9
3	EE6412	Optimal Control	4	0	0	0	8	12
4	EE5340	Micro-electro mechanical systems	3	0	0	0	6	9
5	EE5410	Introduction to DSP	4	0	0	0	8	12
6	EE5177/ CS5011	Machine Learning for Computer Vision	4	0	0	0	8	12
7	EE5175/ CS6350	Image Signal Processing	4	0	0	0	8	12
8	ED5330	Control of Automotive Systems	3	0	0	0	6	9
9	ED5235	Power Electronics and Motor Drives for Electrified Vehicles	3	0	0	0	6	9
10	ED5215	Introduction to Motion Planning	3	0	0	0	6	9

Who are the potential recruiters?

There are several leading companies who actively recruit students with knowledge of robotics. Some of the companies are:

- Eaton Corporation
- Bosch
- TCS Innovation Labs
- Titan
- ABB Robotics
- Kawasaki Robotics
- Pari Robotics
- DRDO
- CSIR
- Systemantics
- HiTech Robotics
- Stryker

Setting up of New Laboratory.

Hands-on experience is a vital component in learning robotics. Keeping this in mind, a laboratory course has been proposed. The laboratory experiments will give practical exposure on various types of robots and their design, programming, and control. The laboratory may be setup in the new academic block to make it accessible to all the faculty involved in robotics teaching/research. Following experiments are proposed in the newly proposed laboratory course.

Experiments:

1. Sensors and Actuators: Integration of various types of sensors, programming, data analysis
2. Serial Manipulators: Forward and Inverse Kinematics, Path Planning, Trajectory tracking, Singularity analysis
3. Parallel Manipulators: Forward/Inverse kinematics, singularity analysis, Path planning
4. Wheeled Mobile Robots: Design, control, motion planning
5. Robotic vision system: Programming, image processing, obstacle detection
6. Underwater robots: Design, modelling and simulation
7. Aerial Robots: design, control, programming (fixed wing/flapping wing/VTOL)
8. Robot calibration: Calibration of industrial robots
9. Surgical robots: Haptic system, RCM design

Facility creation: Investment: 75 lakhs

- Serial robot arm (10 lakhs)
- Parallel robot arm (15 lakhs)
- Mobile manipulator (15 lakhs)
- Haptic system (5 lakhs)
- Mobile robots (10 lakhs)
- Aerial Robot (10 lakhs)
- Robot building kits (10 lakhs)

Conclusions

The structure and curriculum for the interdisciplinary dual degree program in Robotics have been formulated with the objective of leveraging on the academic strengths of each department to ensure the true interdisciplinary nature of the program.

Interdisciplinary Dual Degree in ***tech*MBA**

Vision of the program
To develop human capabilities towards better management of corporations using the technological innovations.
Prepare human resources well-poised to create and manage technology-driven businesses.

Mission of the program: To offer cutting-edge management inputs, with best possible curriculum and industry interface to the participants so that they are capable of managing businesses in the world of newer technologies in the most optimal way.

Learning outcomes for the program:

Category	Learning Outcome (The student will be able…)
Functional Core	To gain a multidisciplinary perspective on business functions – both operational and strategic.
	To develop teamwork and leadership skills in a variety of work group settings.
	To assess opportunities and challenges in domestic and international business contexts.
Performance Analytics	To gain insights into models and tools of business research and management practice.
	To model complex business problems by employing qualitative and quantitative techniques.
	To translate findings of analytics projects into effective and efficient action plans.
Transformation Technologies	To demonstrate deep understanding of the enterprise transformation through digital technologies.
	To design innovative solutions to address business and social problems - using technology.
	To comprehend environmental, cultural, social and ethical dimensions of business decision-making.

Why *tech*MBA?
As the organizations move towards decision making using technological advances, it is imperative that the students learn to marry technology with management. The core management ideas may be evergreen and time invariant; one needs to learn how these can help a technology-based organizations. Also, when so much of technological revolution happening around us, business leaders need to harness and manage these technologies optimally to provide value to all the stakeholders.

techMBA is supposed to help students appreciate principles of how to use technology to manage a business; as well as learn how management principles can help technology organizations.

What is `tech' in *tech*MBA?
The focus of this program is not technology per se, but creating and managing technology-based businesses. We focus on *digital technologies*. Curriculum will have coursework preparing and challenging students to manage digital technologies, basic understanding of these technologies would be pre-

requisites. For, students may need some exposure for some of the emerging digital technologies before they learn ways to manage these. Curriculum will focus on management technologies (e.g. FinTech, Marketing Tech, etc.).

Curriculum

Curriculum can have four broad categories of courses:

1. Core Management Foundations (50%)
2. Performance analytics (25%)
3. Transformation Technologies (25%)

(Figures in bracket indicate approximate proportion of courses.)

THIRD YEAR

Semester 6

No	Course No	Course Name	L	T	E	P	O	C
1	MS4010	Quantitative Techniques for Operations	3	0	0	0	6	9
2	MS4510	Marketing Management: Basics & Application	3	0	0	0	6	9
		Total Credits :						**18**

Summer

No	Course No	Course Name	L	T	E	P	O	C
1		Summer Internship	0	0	0	0	20	0

FOURTH YEAR

Semester 7

No	Course No	Course Name	L	T	E	P	O	C
1	MS4210	Modern Corporate Finance	3	0	0	0	6	9
2	MS4310	Managing People in Organizations	3	0	0	0	6	9
3	MS4410	Information Systems for Organisations	3	0	0	0	6	9
		Total Credits :						**27**

Semester 8

No	Course No	Course Name	L	T	E	P	O	C
1	MS5000	Strategic Management	3	0	0	0	6	9
2	MS4610	Introduction to Data Analytics	3	0	0	0	6	9
3	MSxxxx	Digital Economy and Enterprises	3	0	0	0	6	9
		Total Credits :						**27**

FIFTH YEAR

Quarter 1

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	MS6910	Global Business Management	4	0	0	0	8	6
2	MS5790	Marketing Research	4	0	0	0	8	6
3	MSxxxx	Operations and Supply Chain Analytics	4	0	0	0	8	6
4	MSxxxx	Technology Foresight and Innovatioin	4	0	0	0	8	6
5	MS5015	Design Thinking	4	0	0	0	8	6
		Total Credits :						**30**

Quarter 2

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	MSxxxx	Foundations of Business Analytics	4	0	0	0	8	6
2	MSxxxx	Operations Forensics	4	0	0	0	8	6
3	MS5690	Computational Finance	4	0	0	0	8	6
4	MSxxxx	Digital Business Models	4	0	0	0	8	6
5	MSxxxx	Foundations of Technopreunership	4	0	0	0	8	6
		Total Credits :						**30**

Quarter 3

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	MSxxxx	Capstone Project	0	0	0	0	36	18
2	MSxxxx or MS5700	People Analytics or Derivatives and Risk Management	4	0	0	0	8	6
3	MSxxxx or MSxxxx	Managing Digital Products or Digital Marketing	4	0	0	0	8	6
		Total Credits :						**30**

Quarter 4

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	MSxxxx	Industrial Project	0	0	0	0	24	12

Year	**3rd Year**	**4th Year**		**Summer**	**5th year**				**Total**
Semester / Quarter	**VI**	**VII**	**VIII**		**I**	**II**	**III**	**IV**	
Credits	**18**	**27**	**27**	**0**	**30**	**30**	**30**	**12**	**174**

Interdisciplinary Dual Degree in Quantum Science and Technology (QuST)

The 6 th /7 th Semester B.Tech students are eligible for up gradation to five year Inter Disciplinary Dual Degree (B.Tech & M.Tech) program.

Along these lines, we would like to propose offering a DD Program on Quantum Science and Technology (QuST). Given the widespread interest in quantum computing and information today, both in academia and increasingly in industry, we are hopeful that such a program will attract a fair number of students from EE, MME, ME, CS departments in addition to EP students.

The students have to opt for four electives (36 credits) from the list of electives in advanced materials and nano sciences across different departments

List of courses for DD in Quantum Science and Technology

There will be **four core courses** that include a course on Quantum Computation and Quantum Information, an ID course on Experimental Techniques for Quantum Computation and Metrology, a course on Quantum Electronics and Lasers and a course on Optical Signal Processing and Quantum Communications. (**36 credits** of DD core courses).

The core component also includes a project during the 9th and 10th semesters worth **85 credits.**

Semester VII

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	PH5840	Core 1: Quantum Computation and Quantum Information	3	0	0	0	6	9
2	EE4348	Core 2: Quantum Electronics and Lasers (Includes a 3-credit lab component)	3	0	3	0	6	12
		Total Credits :	**6**	**0**	**3**	**0**	**12**	**21**

Semester VIII

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	ID5843	Experimental Techniques for Quantum Computation and Metrology	3	0	0	0	6	9
2	EE6502	Optical Signal Processing and Quantum Communications	3	0	0	0	6	9
3	Elective I	To be selected from the given list of courses	3	0	0	0	6	9
4	Elective II	To be selected from the given list of courses	3	0	0	0	6	9
		Total Credits :	**12**	**0**	**0**	**0**	**24**	**36**

Semester IX

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	ID5790	Project – I (Summer)	0	0	0	0	25	25
2	ID5791	Project – II	0	0	0	0	20	20

3	Elective III	To be selected from the given list of courses	3	0	0	0	6	9
4	Elective IV	To be selected from the given list of courses	3	0	0	0	6	9
		Total Credits :	**6**	**0**	**0**	**0**	**57**	**63**

Semester X

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	ID5792	Project – III	0	0	0	0	40	40
		Total Credits :	**0**	**0**	**0**	**0**	**40**	**40**

Total credits for the DD program: 160 Credits

(39 Core + 36 Electives + 85 Project)

Elective Courses

(A) Set of relevant electives that are already offered by the **physics** department:

- PH 5842 Advanced Topics in Quantum Information
- PH 5170 Quantum Mechanics – II
- PH 5620 Coherent and Quantum Optics
- PH 5480 Advanced Statistical Physics
- PH 5680 Superconductivity and applications
- PH 5500 Dynamical Systems
- PH 5815 Ultrafast Lasers and Applications

(B) Set of relevant courses (both core and electives) that are already offered by the **EE** department:

- EE5120: Linear Algebra
-EE5142: Introduction to Information and Coding theory
- EE5160: Error control coding
- EE5347: Electronic and Photonic Nanoscale Devices
- EE6500: Integrated Optoelectronic Devices and Circuits
- EE6700: Advanced Photonics Laboratory
- EE7500: Advanced topics in RF and Photonics

(C) Courses from Math and CS departments which can also be a part of the elective list for this program:

- MA5310: Linear Algebra
- CS5011: Introduction to Machine Learning
- CS6111: Foundations of cryptography
- CS7111: Advanced Topics in Cryptography

Students can choose either EE5120: Linear Algebra, or MA5310: Linear Algebra, but not both

Interdisciplinary Dual Degree in Complex Systems and Dynamics

1. Introduction

Complex systems consist of a large number of smaller entities that interact with each other – usually in a nonlinear and stochastic manner. Examples of complex systems can be diverse including, but not limited to, coupled neurons in the brain, spatially distributed interacting systems like the movement of tectonic plates in a planet, transportation network of a large metropolitan, online social networks like Twitter and Facebook, and the ocean-atmosphere coupling in the climate system. Predicting phenomena such as brain epilepsy, climate change and global warming, pandemics, failure of networks involving distributed computing, power or transportation, species extinction in biological ecosystems, earthquakes, bio-mimetic flows and fluid-structure interactions, viral social media posts, forecasting financial crisis, crowd management etc.; these require a deeper understanding of the overall dynamics of the associated multi-physics systems and a complex system approach to investigate it. Mathematical models of such systems are complex, are typically nonlinear and stochastic and involve modeling the coupling between the interacting individual systems. The traditional reductionist approach that builds an understanding of the individual components, is not suitable for analyzing the collective behaviour. This necessitates the development of new techniques and approaches that have come to be known collectively as complex systems approach.

This is a rapidly emerging interdisciplinary field that enables bridging the gap between fundamental knowledge and mathematical models with the empirical observations as well as copious dataset available on account of the advent of new technological innovations that were unthinkable even a few years ago The identification, organization and analysis of such data need to draw on the methods of data mining, artificial intelligence, neural nets and deep learning algorithms. The efficient use of this data in conjunction with the methods of probabilistic and causal analysis can lead to significant improvements in the construction of realistic models, as well as in the prediction of catastrophic events such as earthquakes, fluid structure interaction problems, power grid collapses, stock market collapses, and pandemics of disease that can occur in such systems. The impact of these technical skills has the potential to directly address global challenges arising from human-nature interactions such as, climate change, pandemics, mass migration and ecological disorders, together with the possibility of harnessing technology towards growth and the improvement of life and the environment.

2. Program Objectives

The aim of the proposed program is to introduce students to new techniques and tools for mathematical modelling and analysis of complex dynamical systems and to investigate some of the challenging dynamical problems in climate science, neuroscience, biological systems, Multiphysics systems and active flows, that are the focus of current research worldwide. In addition to enhancing the fundamental understanding of the universal features, which contribute to similar phenomena that occur across a diversity of systems, the effort could also translate into delivering technology which is useful in industrial and societal contexts.

The primary objectives are to train students

- In building network based mathematical models for large scale complex dynamical systems using observational data and use the new theories of complex networks for analysis
- In theories of nonlinear dynamical systems that enables analysis of complex systems which are inherently nonlinear
- In high performance computing skills and data analysis

To achieve these objectives, the course curriculum consists of three compulsory core courses

(a) one from Complex Networks basket that introduces students to complex networks analysis
(b) one from Nonlinear Dynamics basket that exposes students on this subject
(c) one from a select set of Mathematics and Numerical Techniques courses. The courses have been selected to ensure that students can cater to be trained in appropriate skill sets that will be more suitable for their project.

The electives are to be selected from a basket of carefully curated courses that encompass the interdisciplinary nature of the program. The electives can be selected from courses on Data Science, High performance computing, as well as a mix of elective courses applicable to diverse engineering fields, which will enable the students to combine the traditional skills with the new approaches of complex dynamical systems.

Semester VI

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	Core 1 (any one)	Core Basket 1	4	0	0	0	8	12
		Core Basket 2	3	0	0	0	6	9
		Total Credits				**12 or 9**		

Semester VII

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	Core 2 (any one)	Core Basket 2	3	0	0	0	6	9
		Core Basket 1	4	0	0	0	8	12
		Total Credits				**9 or 12**		

Semester VIII

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	Core 3	Core Basket 3	3	0	0	0	6	9
		Total Credits						**9**

Summer

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	ID5890	Project – I	0	0	0	0	20	15
		Total Credits						**15**

Semester IX

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	ID5891	Project – II	0	0	0	0	30	30
		Total Credits						**30**

Semester X

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	ID5892	Project – III	0	0	0	0	40	40
		Total Credits						**40**

Note:
Core Courses – 30 credits
Electives – 45 credits (5 x 9 credit courses or 3x12 credits +1x 9 credit course)
Project – 85 credits
Total IDDD credits - 160

- Core 1 and Core 2 must be completed by Semester 7.
- Core 3 can be credited in Semester 8 in lieu of Semester 9
- Electives are to be completed in Semesters 7, 8 and 9.

Core Basket 1: Networks								
ID XXXX	Complex Networks	4	0	0	0	8	12	New
EE5154	Complex Network Analysis	4	0	0	0	8	12	Existing
CS6012	Social Network Analysis	4	0	0	0	8	12	Existing

*The proposed new course on ID XXXX Complex Networks will be offered by a faculty from this team.

Core Basket 2: Nonlinear Dynamics								
AM5650	Nonlinear Dynamics	3	0	0	0	6	9	Existing
PH5500	Dynamical Systems	3	0	0	0	6	9	Existing
PH5830	Advanced Dynamical Systems	3	0	0	0	6	9	Existing
MA6050	Dynamical Systems	3	0	0	0	6	9	Existing

*At least AM5650 & PH5500 will be offered by faculty members from this team.

Core Basket 3: Mathematics and numerical analysis								
PH5050	Mathematical Physics II	3	0	0	0	6	9	Existing
PH5730	Methods of Computational Physics	3	0	0	0	6	9	Existing
MA5470	Numerical Analysis	3	0	0	0	6	9	Existing
MA6005	Applied Linear Algebra	3	0	0	0	6	9	Existing
MA5014	Applied Stochastic Processes	3	0	0	0	6	9	Existing
MA5312	Stochastic Differential Equations	3	0	0	0	6	9	Existing
MA5890	Numerical Linear Algebra	3	0	0	0	6	9	Existing
MA5892	Numerical Methods in Scientific Computing	3	0	0	0	6	9	Existing
AM5117	Analytical Methods in Engineering Mechanics	3	0	0	0	6	9	Existing
AM5600	Computational Methods in Mechanics	3	0	0	0	6	9	Existing
AS6520	Mathematics for Aerospace Engg	3	0	0	0	6	9	Existing

Electives basket								
MA5013	Applied Regression Analysis	3	0	0	0	6	9	Existing
AM5340	Stochastic processes in mechanics	3	0	0	0	6	9	Existing
AM5630	Foundation of Computational Fluid Dynamics	3	0	0	0	6	9	Existing
AM5116	Structural Control	3	0	0	0	6	9	Existing
AM5450	Fundamentals of Finite Element Analysis	3	0	0	0	6	9	Existing
AM5030	Linear Dynamical Systems	3	0	0	0	6	9	Existing
AM6513	Advanced Computational Fluid Dynamics	3	0	0	0	6	9	Existing
AS6050	Dynamic Fluid Structure Interaction	3	0	0	0	6	9	Existing
AS5850	Finite Element Analysis	3	0	0	0	6	9	Existing
AS5470	Unsteady aerodynamics of moving bodies	3	0	0	0	6	9	Existing
AS6041	Advanced CFD-Eddy Resolving Methods	3	0	0	0	6	9	Existing
BT6270	Computational Neuroscience	3	0	0	0	6	9	Existing
CH5350	Applied Time Series Analysis	3	0	0	0	6	9	Existing
CH6020	Computational Fluid Dynamics Tech	3	0	0	0	6	9	Existing
CH5230	Data driven modelling of Process Systems	3	0	0	0	6	9	Existing
CH6760	Hydrodynamics of Complex Fluids	3	0	0	0	6	9	Existing
MA5013	Applied Regression Analysis	3	0	0	0	6	9	Existing
ME6151	Computational Heat and Fluid Flow	3	0	0	0	6	9	Existing
ID6107	Perturbation Methods for Engineering Problems	3	0	0	0	6	9	Existing
PH5010	Mathematical Physics-1	4	0	0	0	6	10	Existing
XXXXXX	Parallel Scientific Computing	3	0	0	1	6	10	New
BT5240	Computational Systems Biology	4	0	0	0	6	10	Existing
CH5019	Mathematical Foundations for Data Science	4	0	0	0	8	12	Existing
CS5011	Introduction to Machine Learning	4	0	0	0	8	12	Existing
CS5820	Probability and Computing	4	0	0	0	8	12	Existing
CS6023	GPU Computing	4	0	0	0	8	12	Existing
CS6750	Grid Computing	4	0	0	0	8	12	Existing
CS6847	Cloud Computing	4	0	0	0	8	12	Existing
CS6440	Distributed Computing	4	0	0	0	8	12	Existing
CS 6310	Artificial Neural Networks	4	0	0	0	8	12	Existing
CS7015	Deep Learning	4	0	0	0	8	12	Existing

The projects are expected to be on the following areas: climate science, computational neuroscience, machine learning in fluid dynamics, complex flows and physics of living matter, urbanclimate modelling, machine learning in climate science, dynamics of multi-physics systems such us fluid structure interactions, dynamics of social behaviour, causal dynamical networks, stochastic dynamics of spatio-temporal systems etc.

4. Employment opportunities

Students graduating from this program will be proficient in the theories and techniques of modelling and analysis of complex dynamical systems, curating voluminous data for mathematical modelling, high performance computing and will have hands-on experience in applying these techniques for analysis of cutting edge niche problems that encompass fields of electrical engineering sciences, mechanical engineering sciences, biology, physics and mathematics. This is an emerging field and students trained in this area will have opportunities for placement in niche companies and start-ups dedicated to analysis of problems that typically have been difficult for mathematical modelling and analysis.

The Curriculum for the Interdisciplinary Dual Degree (IDDD)- Cyber PhysicalSystems Programme

Preamble

Smart infrastructure and services are essential for developing smart cities and industries.The development of smart services and infrastructure involves integration of multiple fields such as control theory, wireless communication, real-time data analysis, sensors etc. The interactions between the physical, control and communication layers and technologies give rise to networked systems, named as cyber-physical systems**.** Power grid, water-distribution networks, transportation systems, health care etc. are examples ofcyber-physical systems. Design, control, and optimization of cyber-physical systems is essential for developing next generation smart infrastructure and services. The IDDD-Cyber physical systems program is designed to strengthen the participants' skillsin design, control, and optimize cyber-physical systems in theory and practice.

Who offers the programme?

The IDDD- CPS programme is offered by an interdisciplinary group involving faculty members from Electrical engineering, Chemical Engineering, Computer Science andApplied Mechanics.

Who can enrol in this programme?

A B.Tech student from any branch IIT Madras can opt for this IDDD program. The selection to the program will be based on the CGPA at the end of Semester 5.

What is the curriculum?

The IDDD curriculum is designed to cover the fundamentals concepts and tools of disciplines involved in studying CPS. Three core courses provide a background of control theory, mathematical and data sciences, and cyber security. Two labs are designed to provide computational and experimental background in the internet of things, smart infrastructure such as health care, transport and water networks. The students are then free to choose 4 electives from the elective list. These electives are a mix of advanced topics in a control theory, data analysis, IoT, communications etc. The students can choose courses in their areas of interest.

Sr.No.	Course No.	Course Name	L	T	E	P	O	C
Semester 7								
1	Core-1	Core - 1 Basket (Control Theory)	3	0	0	0	6	9
2	Core-2	Core - 2 Basket (Mathematical Science and Data Science)	3	0	0	0	6	12
3	Elective-1							9
4	EE4708	Data Analytics Lab				3	3	6
								36

Sr.No.	Course No.	Course Name	L	T	E	P	O	C
Semester 8								
1	ID5015	Security for Cyber-physical Systems	4	0	0	0	8	12
2	Core-4	Communication networks for IoT	3	0	0	0	6	9
3	Elective-2							9
5	ID5025	IoT/CPS lab				3	3	6
								36

Sr.No.	Course No.	Course Name	L	T	E	P	O	C
Summer								
1	IDXXXX	Project I	0	0	0	0	20	20
								20

Sr.No.	Course No.	Course Name	L	T	E	P	O	C
Semester 9								
1	IDXXXX	Project II	0	0	0	0	30	30
								30

Sr.No.	Course No.	Course Name	L	T	E	P	O	C
Semester 10								
1	IDXXXX	Project III	0	0	0	0	35	35
								35

Sr. No	Semester 7	Semester 8	Summer	Semester 9	Semester 10	Total
1.	36	36	20	30	35	157

Core 1: Control Theory

Sr.No.	Course No.	Course Name	L	T	E	P	O	C
1	CH5120	Modern Control Theory	3	0	0	0	6	9
2	EE5413	Linear Dynamical Systems	3	0	0	0	6	9

Core 2: Mathematical/Data Science

Sr.No.	Course No.	Course Name	L	T	E	P	O	C
1	CH5019	Mathematical Foundations of Data Science	4	0	0	0	8	12
2	EE5412	Mathematical methods in Systems Engineering	4	0	0	0	8	12

Electives

CH5230	Data-driven modelling of Process Systems	9
CH5170	Process Optimization	9
CH5020	Statistical design and analysis of experiments	9
CH5115	Parameter and state estimation	10
EE6432	Stochastic Control	12
EE6433	Distributed Optimization for Control	9
EE6430	Fundamentals of Linear Optimization	9
EE6415	Nonlinear Control Systems	9
EE6412	Optimal Control	9

EE5156	Internet of Things and Management of discrete entities	9
EE5180	Introduction to Machine Learning	12
EE5141	Introduction to Wireless and Cellular Communication	9
CS6650	Smart Sensing for Internet of Things	12
CS6700	Reinforcement Learning	12
CS6330	Digital System Testing and Testable Design	12
AM5140	Biomedical Instrumentation	9
EE6417	Advanced Topics in Control Systems	9

Interdisciplinary Dual Degree in Electric Vehicles (ID-DD-EVs)

The Interdisciplinary Dual Degree programme in EVs has been conceived to cater to the medium-term needs of Automotive OEMs and their Tier 1 vendors- as well as the requirements of the numerous StartUps that are evolving in this space. It is proposed to develop industry-ready professionals who can take up careers in the Functions of Engineering and Development of various types of EVs- starting from e2W and e3W and going all the way to eBuses and eTrucks. The different skills and domain required to train such professionals lie in different departments of IITM- and it is hence imperative that it is offered as an Inter-Disciplinary degree. Upon completion of the degree, the students will be able to, given their strong and application-oriented fundamentals, take up immediate and productive Engineering and Developmental roles. While care has been taken to minimise the number of new courses in the curriculum, existing courses already being offered would be modified marginally to suit the EV domain as well as to increase the practicality/ application of the training by way of Micro Projects and Application Assignments/ Tutorials.

Learning Outcomes:

Students graduating with a dual degree in EVs shall be capable of understanding, conceptualising, analysing, applying and debugging in the following areas related to EVs:

1. Vehicle Dynamics.
2. Battery Engineering, including Cell Development and use.
3. Power Electronics and Embedded Systems for EVs.
4. Motors and their Controllers.
5. Vehicle Control strategies and algorithms.
6. Financial, Market and Economic considerations for an EV EcoSystem.
7. Material Engineering, including advanced Materials and their processes, for EVs.
8. The Fundamental Taxonomy, Topology and Architecture of xEVs.
9. Thermal Management of all EV systems and aggregates.

Who offers the programme?

The ID-DD programme is offered by faculty from the departments of Engineering Design, Mechanical Engineering, Chemical Engineering, Electrical Engineering, Chemistry, Management Studies and Metallurgical & Materials Engineering.

Who can enrol in this programme?

A B. Tech student or a Dual Degree student of IIT Madras in any discipline (except biosciences) is eligible to upgrade/opt for this programme provided the student has a CGPA of 8.0 or above up to 5^{th} semester. Total number of seats will be fixed at 25 and allocation of dual degree specialization and award of the degree will be governed by the rules of the Institute.

What is the future potential for Students who completed ID-DD?

<u>For those students seeking to go in for employment right after this ID-DD:</u>

The curriculum has been drawn up after very detailed inputs were received from Industry. Over time, further industry-oriented courses are to be included with due process and approvals- with members from Industry actually conducting those courses as Adjunct Faculty. The balanced approach to the fundamentals of vehicles, and deep science behind each topic will help the students perform readily after being placed in Industry. Moreover, the Projects and Assignments, along with possible internships, being offered as part of the Program, will help the students be productive assets in employment or as Entrepreneurs right from the first day.

<u>For those students seeking to go in for further studies and research:</u>

The Core as well as the Elective subjects provide for deep dives into the fundamental sciences behind each domain thus equipping the students well for further research or studies, if they so choose. The Electives Basket has been drawn up with focus and care, in order to permit a student to specialise in one of the Domains if he/she so chooses- be it Battery Technology or Power Electronics or Motor Engineering.

What is the curriculum?

ID-DD-EVs has a flexible curriculum. The programme spans a period of five semesters of the five-year dual degree programme. This course will ensure that the students who enter into this specialisation from different streams have the basic understanding of the principles and fundamentals for the different EV-related topics. Some of the subjects, in order to increase the application strengths of the students and to make them more industry-ready, will have Micro Projects (some of them lab-based, where required) as part of the training and assessment. The curriculum also allows short term (1-3 months)/ long term (up to 6 months) internships with potential companies.

In tune with the overall structure of the dual degree program being offered in the Institute, the number of courses to be offered and the credit distribution are as follows:

Total Credits required:	**155 to 160**
No. of PMT CORE courses to be offered:	4 (39 credits)
No. of electives to be offered:	4 (~36 credits)
No. of labs. to be offered:	1 (3 credits, already included in the 39 CORE credits)
Project work/internship	3 (20 + 30 + 35 = 85 credits)

Total credits for the ID-DD specialization: ~160 credits

Interdisciplinary DD in EVs -course curriculum

Sl. No	Course No	Course Name	L	T	E	P	O	C
Semester 6								
C1	ED5220	Core 1: Vehicle Dynamics	3	0	0	3	6	12
C2	ED5235	Core 2: Power Electronics and Drives for Electrified Vehicles	3	0	0	0	6	9
		Total Credits :						21
Semester 7								
C3	ED5330	Core 3: Control of Automotive Systems	3	0	0	0	6	9
C4	ID5500	Core 4: Battery Technology	3	0	0	0	6	9
		Total Credits :						18
Semester 8								
		Summer						
P1		Project I – Summer Project						20
Semester 9								
P2		Project II						30
Semester 10								
P3		Project III						35

Project: **85** credits to be completed in the summer after the 8th semester, 9th and 10th semesters.

Electives: Around **36** credits to be completed from the approved list in 8th, 9th and 10th semesters.

Total credits for the DD programme: ~160

ELECTIVE COURSES

Faculty/Department consent has been received for the electives or is in process.

		L	T	E	P	O	C
ED5340	Data Science: Theory and Practice	3	0	0	3	6	12
ED5160	Fundamentals of Automotive Systems	4	0	0	3	8	15
ED5515	Fundamentals of Thermal Management in Electric Vehicles	3	0	0	0	6	9
ED5270	Motorcycle Dynamics	3	0	0	0	6	9
CY6015	Electrochemistry: Fundamentals and Applications	3	0	0	0	6	9
CY6998	Electrochemical Approaches to Functional Supramolecular Systems	3	0	0	0	6	9
ME5228	Engineering Acoustics	3	0	0	0	6	9

ID5020	Multi-Body Dynamics and Applications	3	0	0	0	6	9
EE5200	Power Converter Analysis and Design	3	0	0	0	6	9
EE5201	Modelling and Analysis of Electric Machines	3	0	0	0	6	9
EE5203	Switched Mode Power Conversion	3	0	0	0	6	9
EE6262	Advanced Motor Control	3	0	0	0	6	9
EE5204	Electric Vehicles and Renewable Energy	3	0	0	0	6	9

Note: Students are required to take the appropriate pre-requisites for courses or obtain Consent of Teacher.

Interdisciplinary Dual Degree Quantitative Finance

Objectives of the program

- Enable students to more easily adapt to new developments in finance and bridge the gap between application of modern product and process technologies and state-of-the-art finance.
- Build advanced knowledge of the main theoretical and applied concepts in quantitative finance, financial engineering and risk management, using current issues to stimulate the thinking process.
- Prepare for careers involving the design and management of new financial instruments, the development of innovative methods for measuring, or predicting and managing risk.

Eligibility:
Open to all branches.
Minimum CGPA of 8.0 at the end of 5th semester.
Selection will be strictly based on CGPA till the 5th semester.

Credit Requirements

- Core: 60 credits
- Elective: 30 credits
- Project: 70 credits: Earned over the 9th and 10th semesters plus the summer preceding or succeeding them. Can be done in collaboration with industry.
- Total = 160 credits

Course categories

- ***Fundamental Finance*** – To understand the core principles, theories and concepts of finance
- ***Advanced financial tools and techniques*** – To understand the current state-of-the art in the financial markets. Particular focus on the new innovations in fintech.
- ***Computational/Analytical capabilities*** – To acquire analytical skills in the context of finance.

Sr #	***Course #***	***Course name***	***L***	***T***	***E***	***P***	***O***	***C***
Semester 6		Jan – May						
1	MS6621(New)	Corporate Finance	4	0	0	0	8	12
Semester 7		July – Nov						
1	HS5701	Microeconomics -I (if not credited earlier)	3	0	0	0	6	9
2	MA5015	Stochastic Calculus for Finance	3	0	0	0	6	9
3	IDxxx(New)	Introduction to Machine learning	4	0	0	0	8	12
Semester 8		Jan – May						
1	MA 5950	Mathematical Finance	3	0	0	0	6	9
2	MSXXX(new)	Financial Derivatives and Markets	4	0	0	0	6	9
3	IDXxx	Elective 1	4	0	0	0	8	12

Summer								
1	Xxx	Project – Phase 1	0	0	0	0	0	15
Semester 9		July – Nov						
1	xxx	Elective 2	4	0	0	0	8	6
2	xxx	Elective 3	4	0	0	0	8	6
3	xxx	Elective 4	4	0	0	0	8	6
4	xxx	Project – Phase 2	0	0	0	0	0	20
Semester 10		Jan – May						
1	xxx	Project – Phase 3	0	0	0	0	0	35

Note: Students are requested to do the HS3002 Principles of Economics (if not credited earlier) either in the sixth or eight quarter

Proposed elective courses are listed below:
DoMS electives:

- Asset Pricing (MS5617) – 6 credits
- Fixed Income Securities, Structure and Trading (MS6680) – 6 credits
- Financial Risk Management (MS6710) – 6 credits
- Computational Finance (MS5690) – 6 credits
- Venture Capital and Entrepreneurial Finance (MS6640)- – 6 credits
- FinTech and Entrepreneurship [New elective] – 6 credits
- Algorithmic Trading [New elective]- – 6 credits
- Behavioural Finance and Financial Crises [New Elective]-6 credits
- Quantitative Data Science/Analysis in Finance [New Elective] -6 credits
- Quant Finance Seminar [New Elective] – 6 credits
- Dynamic Optimization for Finance [New Elective]-6 credits

IDDD electives

- Machine Learning and Data Science for Finance [New elective]-12 credits
- Time Series Modelling for Finance [New Elective]- 12 credits
- Block Chain technology and Crypto currency [New Elective]- 12 credits
- Deep learning Techniques [New Elective]- 12 credits

*MSXXX(new) Financial Derivatives and Markets, IDxxx(New) Introduction to Machine learning -The syllabus for these courses will be sent for approval to BAC subsequently as they will be offered in July-Dec2023/Jan-May 2024.

Interdisciplinary Dual Degree in Public Policy

To be offered by the Department of Humanities and Social Sciences in Collaboration with Engineering and Management Departments

Motivation / Rationale for this Programme:

Policy making processes in the future, as in the past, will require a sound understanding of technologies that become a part of policy regimes in various sectors. Students of engineering trained in policy sciences will be able to contribute more effectively in the designing, implementation, monitoring and assessment of policies at societal level. The benefits and consequences of new technologies can be realised only if we fully understand their impact on markets, society, citizen rights and regulatory practices. The proposed programme in Public Policy intends to impart such perspectives and skills to the Undergraduate Engineering students of IITM and equip them to make significant contributions to the policy making process in India.

The proposed programme will produce a new breed of professionals trained in Engineering and Policy Sciences. Students of this programme will have opportunities to learn through internships and field projects and engagement with policy making machineries such as Planning Commissions of various state governments, Think Tanks, reputed Non-Governmental Organizations known for their contributions in the past as well as elected representatives (MPs and MLAs).

IIT madras is uniquely placed as a premiere engineering institution with decades of robust training with a vibrant Humanities and Social Sciences faculty that offers the context, political culture and social milieus to make sense of engineering education and what it means in a changing world that is increasingly globalised. It is common knowledge that the problems facing the country and the world today (such as climate change, energy crisis, malnourishment, digital divide among others) require a multidimensional approach and involves training that is interdisciplinary and policy oriented.

Who offers this programme?

Leveraging the availability of resources from the faculty of Humanities and Social Sciences, with support from some other departments such as DoMS, CE, AM, EE, CSE, ED, CH, ME, an innovative interdisciplinary Dual Degree program is proposed for UG engineering students of the Institute. We believe that engineering students with grounding in their respective disciplines and trained in policy studies will add value not only to the students' career prospects, but will also address the needs of the market and central/state governments.

Eligibility:

Students already enrolled in engineering departments of the Institute and are in their 7th semester can register for the programme. To be eligible for this programme, the student should have 8 CGPA at the end of 5th semester. Up to 20 students may be admitted to this programme.

Structure of the Programme and Salient Features:

The core courses are carefully chosen to introduce the students to the discourse of governance, institutions, policy related issues that are socially and economically grounded while locating them in technology, intellectual property rights etc. It then expands to deeper and wider issues in the form of elective courses dealing with health, energy, planning, environment, pollution etc. and also includes an active engagement with macroeconomics, resource economics, data analytics, game theory among others. The programme will provide a through grounding on Governance and Institutional mechanisms in the Indian context, besides skills for assessment of policies and public programmes.

- The programme consists of 5 core courses, 3 electives, and Project work (in three stages).
- Electives may be chosen from a basket of 15 courses (more electives may be added as and when it is required)
- Core courses are offered by the Department of Humanities and Social Sciences and the electives from multiple Departments.
- The programme commences from the 7th Semester with a teaser course titled ‘HS 3031: Technology and Public Policy’.
- The program involves a summer internship (after the 8th Semester).
- To be eligible for the degree, the student is required to complete a minimum of 157 Credits.

Semester wise courses:

Semester VII

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	HS3031	Technology and Public Policy	3	0	0	0	6	9
		Total Credits						**9**

Semester VIII

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	HS5950	Economic Analysis of Public Policy	3	0	0	0	6	9
2	HS5952	Governance and Institutions in India	3	0	0	0	6	9
3		Elective - 1						9
		Total Credits						**27**

Summer

Sl. No	Course No	Course Name	L	T	E	P	O	C
1		Project 1	0	0	0	0	20	20
		Total Credits						**20**

Semester IX

Sl. No	Course No	Course Name	L	T	E	P	O	C
1	HS5550	Research Methodology	3	0	0	0	6	9
2	HS5953	Pollution, Public Health and Public Policy	3	0	0	0	6	9
3		Elective - 2						9
4		Elective - 3						9
5		Project 2						20
		Total Credits						**56**

Semester X

Sl. No	Course No	Course Name	L	T	E	P	O	C
1		Project 3						45
		Total Credits						**45**

Total Credits: 157

List of Electives:

Sl No	Course Code	Course Title	Credits
1	ID5060	Energy Economics	9
2	HS3200	Macroeconomics-1	12
3	HS4007	Computational Economics and Public Policy	9
4	HS4330	Environmental and Resource Economics	12
5	HS5125	Global Health and Policy	12
6	HS5709	Indian Economy	9
7	HS5880	Corruption and Development	12
8	HS5951	Intellectual Property Rights: Global and Indian Perspectives	9
9	HS5954	Public Policy for Urban Development	9
10	MS3540	Introduction to Game Theory	9
11	MS4610	Introduction to Data Analytics	12
12	MS4710	Digital Economics and Enterprises	9
13	CE5870	Infrastructure Planning and Management	9
14	CS6012	Social Network Analysis	12
15	ME6148	Renewable Energy Technology	9

Interdisciplinary Dual Degree in Atmospheric and Climate Sciences (ID-DD-ATMOSS)

Program objective:

To provide detailed understanding about atmospheric processes and their interactions covering the range from molecular to global scale helping in improved knowledge about climate change.

Atmospheric and Climate Sciences is an interdisciplinary dual degree programme aimed at nurturing and developing the next generation of atmospheric and climate scientists**.** The effects of climate change are imminent, with far-reaching consequences in all spheres of our life. Increased participation in research related to atmospheric and climate sciences is imperative to improving our understanding of the complex systems involved in climate change. To train and orient potential future researchers toward world-class atmospheric and climate science research programs, we propose an interdisciplinary dual degree master's program to combine the disciplines of science, engineering, and social sciences. The goals of atmospheric and climate science research are to understand, record ultimately, and predict the underlying processes inducing the climate change in an attempt for us to manage, adapt, and help in the design of remedial measures. The Indian climate system is unique in weather patterns and atmospheric sciences, which requires a detailed understanding and focus on research and development in this area to elucidate futuristic climate change impacts. Research and capacity building in this domain need to be backed by a robust fundamental understanding, developed, and supported by local experts, and we need to aggressively reduce our dependence on international agencies. Such dependence should be minimum for need-based collaborations. This program aims to mitigate this status in some measure. IIT Madras has faculty members working in atmospheric sciences and climate change in various departments for quite some time, in their own individual capacities. Atmospheric and Climate Change is a multidisciplinary area, and no single department can have the critical mass to offer a dual degree program in atmospheric sciences; therefore, this proposed interdisciplinary dual degree program will cut across various departments and institutions. The dual degree program in Atmospheric and Climate Sciences will focus on the fundamental elements of atmospheric and associated environmental systems and the technical aspects of climate change at multiple scales of the planetary ecology. The curriculum has been developed with this focus.

Learning Outcomes:

Students graduating with a dual degree in Atmospheric and Climate Sciences should be capable of understanding and applying the following:

1. Fundamentals of Atmospheric Physics and Chemistry
2. Momentum, Energy, and Mass Transport Processes in the Atmosphere
3. Mathematical modelling of Atmospheric systems and climate dynamics
4. Aerosol Science and Technology
5. Environmental Impact and Risk Assessment
6. Measurement, Analysis, and Interpretation of Atmospheric Characteristics
7. Indian monsoon system: Impact of climate change on extreme weather events
8. Impact of climate change on society: Culture and languages

Who offers the program?

The ID-DD program is offered by faculty from the Civil Engineering, Applied Mechanics, Chemical Engineering, Chemistry, Mechanical Engineering, and Humanities and Social Sciences departments. In the future, we expect to integrate faculty members from other departments involved in related research by way of course offerings and guidance of student projects.

Who can enrol in this program?

A B. Tech student or a Dual Degree student of IIT Madras in any discipline can upgrade/opt for this program provided the student has a CGPA of 8.0 or above up to the 5^{th} semester. The total number of seats will be fixed at 10, and the rules of the Institute will govern allocation of dual degree specialization and award of the degree.

What is the curriculum?

ID-DD-ATMOSCC has a very flexible curriculum. The programme spans four semesters of the five-year dual degree programme. This course will ensure that the students who enter this specialisation from different streams have a basic understanding of atmospheric and climate systems to equip them to pursue further in-depth research or application in any area of atmospheric and climate sciences. This program can also orient students appropriately to the opportunities available in the current landscape of entrepreneurship.

L: Lecture; **T:** Tutorial; **E:** Extended Tutorial; **P:** Laboratory; **O:** Outside class hours; **C:** Credits

Project: 85 credits to be completed in the Summer (after 8th Semester), 9th and 10th semester

Electives: 3 electives to be completed from the basket of elective courses given below.

In tune with the overall structure of the dual degree program being offered in the Institute, the

Sr.	**Course Number**	**Course Name**	**L**	**T**	**E**	**P**	**O**	**C**
	Semester 7							
1	ID5127	Core 1: Introduction to Atmospheric Science	3	0	0	0	6	9
2	CE5235	Core 2: Understanding climate dynamics and its mysteries	3	0	0	0	6	9
3	CEXXX	Atmospheric Science Laboratory	0	0	0	6	3	9
		Total Credits						**27**
	Semester 8							
1	CE5525	Core 3: Atmospheric Chemistry and Physics	3	0	0	0	6	9
2		Elective 1						9
3	CE5XXX	Core Lab 1: Atmospheric Simulation Laboratory	0	0	0	6	3	9
4		Elective 2						9
5		Total Credits						**36**
		Project-I (Summer)						**20**
	Semester 9							
1		Elective 3						9
2		Project 2						25
		Total Credits						**34**
	Semester 10							
1		Project 3						40
		Total Credits						**40**
							Total Credits	**157**

Number of courses to be offered and the credit distribution are as follows:
No. of PMT CORE courses to be offered: 3 (27 credits)
No. of electives to be offered: 3 (27 credits)
No. of CORE laboratory courses to be offered: 2 (18 credits)
Project work/internship: 1 (85 credits)

Total credits for the ID-DD specialization: 157

Structure of the laboratory courses: The laboratory courses are designed in such a way that each laboratory experiment/simulation will have one hour of theory classes to orient student with the detailed fundamentals about the experiments and model, which is being covered in the laboratory class. For example, the simulation laboratory, which will introduce students to the global climate models like WRF-Chem, GEOS-Chem, etc. will also include one hour of theory about the development, governing equations, and structure of the models. This will be followed by the hands on the global models, which need to be run on large clusters. Some of the course modules and laboratory experiments will be co-taught by our collaborators from Harvard University and Max Planck Institute for Chemistry.

ELECTIVE COURSES

Electives will be offered as per the following list of the electives. Students are required to take three electives from the following list. Consent from the respective Faculty members /Dept. have been received for all the electives.

	Course No	Course Name	L	T	E	P	O	C1
1	CH5370	Environmental Quality Monitoring and Analysis	3	0	0	0	6	9
2	CE6180	Environmental Impact Assessment	3	0	0	0	6	9
3	CE5971	Aerosol Science and Technology	3	0	0	0	6	9
4	CYXXXX	Atmospheric Chemistry and Processes	3	0	0	0	6	9
5	CE5015	Environmental Monitoring and Data Analysis	3	0	0	0	6	9
6	CE5260	Models for water and Air Quality	3	0	0	0	6	9
7	ID5025	Geophysical Fluid Dynamics	3	0	0	0	6	9
8	CH5350	Applied Time Series Analysis	3	0	0	0	6	9
9	ID5035	Climate Change and Society	3	0	0	0	6	9
10	CE6215	Soil-Plant-Atmosphere continuum	3	0	0	0	6	9
11	CE5960	Remote Sensing of Earth Resources	3	0	0	1	6	10

This list of electives will be updated time to time based on availability of new courses relevant to this program being offered in the future, with the necessary approval of the respective DCCs, BAC, and the senate.

Potential Career Options

- Ph.D. in Atmospheric and Climate Sciences Program all over the World. Nominally there are more than 100 PhD positions are advertised per year all across the globe in atmospheric and climate sciences, and allied areas. With increasing awareness about climate change impacts, a demand for climate scientists is also on increase specifically about Indian climate system.

- Organisations working on Environment, Weather, Climate Change, Ecology and Energy Issues (E.g. TERI, IMD (Indian Meteorological Department, WMO (World meteorological Organisation), UNEP (United Nations Environment Program), CPCB (Central Pollution Control Board), Climate Investment Funds (CIF), GCF (Green Climate Fund))
- Industries manufacturing meteorological/climate/environmental sensors.
- Non-Governmental Organisations, consultancy firms, and think tanks working in the areas of climate sustainability, climate policies, climate change and risk assessment, etc.

Infrastructure: The major infrastructure required for this program is adequately and sufficiently available at IIT Madras, which include but not limited to advanced state-of-the-art particle and gas phase instruments, weather stations, and supercomputing facilities.

Conclusions:

At the end of this program, the students will be well versed with the fundamentals and underlying processes of complex atmospheric system with special emphasis on Indian climate. Students are expected to improved opportunities for higher studies in the top ranked universities and institutions across the globe in atmospheric and climate sciences. This program will help in achieving the following goals:

- Provide students the quantitative assessment of atmospheric dynamics, climate change factors and interactions, and societal impacts
- In-depth understanding and practical knowledge about numerical and climate models
- Hands on experience on atmospheric chemistry and physics laboratory, and
- Participate in real-time and on ground large scale international field measurement campaigns with top collaborators.